ERS O&M Portal — Data Dictionary

Updated 14 October 2026. Written from `server.py` and a real `db.json`, and checked against them. This replaces the version dated 3 October, which is out of date (it missed engineer-only visits, Format B reports, NTP checklists and several fields).

1. Start here

All data lives in one file, `db.json`. There are no other tables. The server loads the file, changes it in memory and writes the whole file back on every change. If you move to a real database, each section below becomes one table (or one table plus child tables).

Three things to know before you read on:

1. **People are stored by name, not by id.** A ticket stores `"client": "Acme Sdn Bhd"` and `"assignees": ["CS - Alice"]`. A client or account must therefore never be renamed. In a real database use ids.
2. **The report lives inside the ticket.** There is one report per ticket, and it has no separate table.
3. **Photos and files are only text labels.** No file is stored anywhere.

2. How to read the tables

Word	Meaning
text	A string. Empty text `""` means "not filled in".
date	Text in the form `YYYY-MM-DD`. This is a Malaysian calendar date.
timestamp	Text such as `2026-10-14T09:58:07+00:00`. Always UTC. Add 8 hours for Malaysia.
list	A JSON list `[...]`
object	A JSON object `{...}`
label	A text standing in for a photo or file (a mock upload).

Word	Meaning
clean	All text typed by users has `<`, `>` and `"` replaced by `<`, `>` and `"` before saving (not Knowledge Base text).

3. The top of db.json

Key	What it holds	Notes
`_seq`	Last ticket number used	Next ticket is `TCK-` plus this number plus 1, four digits. Numbers are never reused.
`_client_seq`	Last client number	`CLI-0001` style
`_calendar_seq`	Last calendar event number	`EVT-0001` style
`_report_seq`	Last report number	Used in `RPT-YYYYMMDD-NNNN`
`_kb_article_seq`	Last article number	Appears only after the first article is made. `KB-0001` style.
`settings`	Global settings	Section 10
`accounts`	The 9 demo users	Section 9
`clients`	Customers	Section 8
`calendar_events`	Events that stand in for Google Calendar	Section 8
`tickets`	Every ticket, newest first	Sections 4 to 7
`kb_articles`	Knowledge Base articles	Appears only after the first article is made. Section 10.

4. Ticket

One entry in `tickets`. The ticket is the centre of the system.

Field	Type	Meaning
`id`	text	`TCK-0001` style.
`title`	text	If left empty at creation: client name plus type name.
`client`	text	Client **name**. Must match a client when created.
`site`	text	Free text.
`type`	text	See the type table in section 11.
`priority`	text	`low`, `medium`, `high`, `critical`.
`description`	text	What the customer reported or needs.
`state`	text	`created`, `scheduled`, `site_visit`, `report_drafting`, `engineer_review`, `approved`, `closed`.
`assignees`	list of names	HOD, Engineer or CS only, never a technician. The creator is first.
`default_engineer`	text	Set at first scheduling. Used for new visit days.
`engineer_date`	date or null	The date of the earliest visit day. Kept in step automatically.
`visit_days`	list	Section 5.
`calendar_event_ids`	list	Events linked to this ticket.
`report`	object	Section 6.
`closure`	object	`type` (`resolved` or `early`), `reason`, `closed_at`. All empty until closed.
`snooze_until`, `snooze_reason`, `snoozed_at`	date, text, timestamp	Hides the ticket for a while. Empty if not snoozed.

Field	Type	Meaning
`reschedule_requested`	true or false	Only present after the first request. While true no checklist can be submitted.
`reschedule_reason`, `reschedule_requested_at`	text, timestamp	Only present after a request.
`messages`	list	The ticket thread. See below.
`mention_reads`	object	Account name to the number of messages that person has seen. Drives the bell icon.
`equipment`	object	`panels`, `inverter_brand`, `inverter_model`, `nemcd_ref` (all text).
`commercial`	object	`dlp_end_date` (text) and `chargeable_service_call` (true or false).
`additional`	object	`so_number`, `premium_care_status`, `system_size`, `assigned_team`, `contact_phone`, `client_remarks` (all text).
`event_log`	list	The audit trail. See below.
`created_at`, `updated_at`	timestamp	

Message (in `messages`)

Field	Meaning
`id`	Number: 1, 2, 3... in order
`author`	Account name
`ts`	Timestamp
`text`	The message
`mentions`	List of account names that were @mentioned

Event log entry (in `event_log`)

Every successful change adds one entry. Nothing is ever edited or removed, and a correction to a time is stored beside it, not over it.

Field	Meaning
`ts`	Timestamp
`actor`	Account name, or `system` for automatic changes
`action`	What happened. Examples: `create`, `link_events`, `checklist_started`, `checklist_submitted`, `site_visit_started`, `day_unfinished`, `auto_advance`, `report_submitted`, `report_approved`, `report_rejected`, `close`, `early_close`, `snooze`, `reschedule_request`. Report edits are logged too, such as `report_section` and `report_media_add`.
`from`, `to`	The ticket state before and after. Equal when the action does not change state.
`note`	Short extra text, such as a reason or a date

5. Visit day and checklist

A visit day is one calendar date on which someone goes to site. When a calendar event covers three dates, linking it makes three visit days. Each has its own checklist. Each is one entry in the ticket's `visit_days` list.

Field	Type	Meaning
`id`	number	1, 2, 3... within this ticket.
`date`	date	The day of the visit.
`source_event_id`	text	The calendar event it came from.
`engineer`	text	Engineer for this day.
`technicians`	list of names	Technicians for this day. **Empty means an engineer-only visit**: the engineer goes alone and fills the checklist.
`status`	text	`pending`, `submitted` or `unfinished`. `unfinished` is set automatically when the date has passed without a submitted checklist.
`checklist`	object	Below.

Checklist (inside each visit day)

Field	Type	Meaning
`template`	text	`residential`, `ci` or `ntp`. Chosen from the ticket type (section 11).
`checklist_started_at`	timestamp or null	Set when it is first opened.
`checklist_completed_at`	timestamp or null	Set on submit. After this the checklist cannot be changed.
`submitted_by`	text or null	Account name.
`general_remarks`	text	Remarks for the whole day.
`general_media`	list of labels	Photos or voice notes for the whole day.

Field	Type	Meaning
`pdf_file`	text or null	NTP only: the one uploaded PDF name.
`items`	list	The questions. Empty for NTP.

Checklist item (in `items`)

Field	Meaning
`id`	`item-1`, `item-2` …
`stage`	`pre`, `during` or `post`
`section`	The group heading shown on the page
`text`	The question. `**word**` marks a word to show in bold.
`status`	`pass`, `fail`, `na`, or null if not answered
`note`	Text. **Required** when `status` is `fail`, unless there is an attachment.
`attachments`	List of labels

The residential template has 20 questions (5 pre, 13 during, 2 post). The C&I template has 22 (4 pre, 10 during, 8 post). The NTP template has none; it uses `pdf_file`.

What can be changed and when: answers can only be given on the visit date itself (Malaysia time), and never after submit. Submitting needs every question answered, or the PDF for NTP.

6. Report

Stored in `ticket.report`. It is created together with the ticket, and `status` goes `draft`, then `submitted`, then `approved`. A reject sends it back to `draft`. While `submitted` or `approved` its content cannot be edited.

There are two layouts. `format` says which one.

6.1 Fields in both formats

Field	Type	Meaning
`id`	text	`RPT-20261014-0109` style: date made plus a running number.
`format`	text	`residential` (Format A) or `troubleshooting` (Format B).
`status`	text	`draft`, `submitted`, `approved`.
`submitted_by`, `approved_by`	text	Account name. Empty text until it happens.
`submitted_at`, `approved_at`	timestamp or null	
`report_rejections`	list	One entry per reject: `reason`, `actor`, `at`.
`installation_improvement`	object	`site_improvement_performed` (`yes`, `no` or null) and `engineer_comments`.
`additional_media`	list of labels	Extra photos.
`duration_overrides`	object	Only present after the HOD corrects a time: `report_started_at` and `report_submitted_at`, each with `value`, `reason`, `by`, `at`.
`created_at`, `updated_at`	timestamp	

A ticket with an old report that does not match its type is rebuilt automatically when it is read.

6.2 Format A — residential (every type except the two troubleshoot types)

Field	Meaning
`general`	`date_of_inspection` (date, copied from the first visit), `system_owner` (the client name), `system_location` and `contact_no` (copied from the client), `system_size_kwp`, `inverter_model`, `inverter_sn`

Field	Meaning
`maintenance_checklist`	16 checks, each `id` (`chk-1` … `chk-16`), `category`, `name`, `extra_label`, `result` (`accept`, `reject` or null), `remarks`, `extra`. Categories: PV Module & Structure (3), PV Inverter (5), AJB, Trunking, Earth Chamber (4), PV Meter (4). `chk-6` has extra label "Data Period" and `chk-7` has "Ver."
`electrical.ac_phase_type`	`single`, `three` or empty
`electrical.ac_measurements`	`single`: `line_to_neutral_v`, `line_to_ground_v`, `neutral_to_ground_v`. `three`: `l1_l2_v`, `l2_l3_v`, `l3_l1_v`, `l1_n_v`, `l2_n_v`, `l3_n_v`, `l1_g_v`, `l2_g_v`, `l3_g_v`, `n_g_v`. All text.
`electrical.inverters`	List of inverters. Each has `id`, `label`, `mppts`. Each MPPT has `id`, `label`, `strings`. Each string has `id`, `label`, `voltage_v`, `pos_to_ground_v`, `neg_to_ground_v`, `status`.
`media`	Six lists of labels: `pv_before`, `pv_after`, `inverter`, `ac_photo`, `dc_photo`, `fit_meter`
`summary`	`psh`, `overall_status` (`good`, `fair`, `poor`), `yearly_generation_kwh`, `avg_monthly_generation_kwh`, `engineer_comments`, `performance_graph` (list of labels)
`report_notes`	`performance_report_result` (`accept`, `reject` or null), `remarks`, `additional_notes`

6.3 Format B — troubleshooting (`residential_troubleshooting` and `ci_troubleshooting`)

Field	Meaning
`checklist_info`	`owner` (client name), `system_capacity`, `contact_no`, `inverter_model_sn`, `panel_model`

Field	Meaning
`intake`	`date_report_received`, `time_report_received`, `received_by`, `date_attended_site`, `time_attended_site`, `affected_components`. `date_attended_site` is filled from the first visit and is required to submit.
`inspection_checklist.ac_three_phase`	Ten readings: `red_yellow`, `red_blue`, `yellow_blue`, `red_neutral`, `yellow_neutral`, `blue_neutral`, `red_ground`, `yellow_ground`, `blue_ground`, `neutral_ground`
`inspection_checklist.ac_single_phase`	Three readings: `live_neutral`, `live_ground`, `neutral_ground`
each reading	An object with `value` (text) and `remark` (`ok`, `not_ok` or null)
`inspection_checklist.dc_strings`	List, at least one. Each has `id`, `label`, `voltage_v`, `positive_ground_v`, `negative_ground_v`, and a `..._remark` for each (`ok`, `not_ok` or null).
`inspection_checklist.line_items`	Seven fixed items, each `id`, `label`, `value`, `remark`: Solar Panel Visual Inspection, Grounding Inspection, AC and DC Junction Box Inspection, Yield PSH, Capture Irradiance, IV Curve Test, Insulation Test
`reported_issues`, `root_causes`, `actions_taken`, `conclusions`, `remarks`	Five lists of lines. Each line: `id` (number), `text`, `photos` (labels, at most 10).
`signatures`	`customer_name`, `ers_pic`
`site_photos`	List of labels

7. Time and dates

Item	Rule
Timestamps	UTC with `+00:00`. Convert to Malaysia time (UTC+8) when showing.
Visit dates and `engineer_date`	Plain dates for the Malaysian calendar day.
"Today"	Malaysian date. The server uses `BUSINESS_UTC_OFFSET_HOURS` in `server.py` (setting `OM_TZ_OFFSET_HOURS`), and the pages use the same value in `static/shared.js`. **Both must be changed together.**
Day rollover	A visit day opens at 00:00 and closes at 24:00 Malaysian time.

8. Clients and calendar events

Client (in `clients`)

Field	Meaning
`id`	`CLI-0001` style
`name`	Must be unique (capital letters ignored). **Tickets point to this name.**
`type`	`individual` or `company`
`contact_person`, `phone`, `email`, `address`, `city`, `state`, `postcode`	Text, not checked for format
`sites`	List of sites. Each has name, capacity, panels, inverters. The client page and the ticket form read it. No API call adds a site: new clients start empty, so sites exist only in the demo data. This needs a way to add them.
`created_by`	Account name

Field	Meaning
`created_at`, `updated_at`	Timestamps

Calendar event (in `calendar_events`)

Field	Meaning
`id`	`EVT-0001` style
`title`	Text
`date`, `end_date`	Dates. `end_date` equals `date` for a one-day event.
`all_day`	true or false
`time`, `end_time`	`HH:MM`, or empty text for an all-day event
`linked_ticket_id`	The ticket it is linked to, or null. **The ticket's `calendar_event_ids` must agree with this.** An event can belong to only one ticket.
`created_at`	Timestamp

In the real system, events should arrive from Google Calendar. Today a form on the Calendar page makes them.

9. Accounts

In `accounts`. There is no password and no email.

Field	Meaning
`id`	`acc-hod-1`, `acc-eng-1`, `acc-cs-1`, `acc-tech-1` …
`name`	Display name such as `CS - Alice`. This is also the key used everywhere else.
`role`	`admin` (shown as HOD), `engineer`, `cs`, `technician`
`team`	Technicians only: `Team 1` or `Team 2`. Decides which visit days a technician may fill.

10. Settings and Knowledge Base

settings

Field	Meaning
`stale_minutes`	A number. Used by the pages to flag a ticket that has not moved.

Knowledge Base article (in `kb_articles`)

Field	Meaning
`id`	`KB-0001` style
`title`	Required
`category`	`Customer Guides`, `General Knowledge`, `Intern Resources`, `Flowcharts` or `Emergency Response`
`visibility`	`internal`, `intern_admin`, `customer`, `public`. The server only stores this; the page decides what to show.
`body`	Text. Not cleaned on input, so escape it when you show it.

Field	Meaning
`status`	`draft` or `published`. New articles are always `draft`.
`attachments`	List of file name labels, at most 10
`created_by`	Account name
`created_at`, `updated_at`	Timestamps

The pages also hold some built-in articles in the page code. Changing the audience of one of those is not saved.

11. Fixed value lists

Ticket type and what it gets

`type`	Shown as	Checklist	Report
`residential_servicing`	Residential Servicing	residential	Format A
`residential_troubleshooting`	Residential Troubleshoot/Rectification	residential	Format B
`ci_servicing`	C&I Servicing	ci	Format A
`ci_troubleshooting`	C&I Troubleshoot/Rectification	ci	Format B
`site_survey`	Site Survey	ci	Format A
`dere`	DERE	ci	Format A
`ntp_phase_1` to `ntp_phase_4`	NTP Phase 1 to 4	ntp (PDF)	Format A
`servicing`, `troubleshooting`	Servicing, Troubleshooting	residential	Format A

The last row is the **old** pair. It cannot be chosen any more, but about half of the demo tickets still carry it. They get the residential report, even `troubleshooting`. Move them to a new type when you migrate.

Note: the HOD confirmed on 14 October that Residential Troubleshoot/Rectification uses the residential checklist questions with the Troubleshoot (Format B) report, as in the table above. This matches the code, so nothing needs to change.

Other lists

Where	Allowed values
`ticket.state`	`created`, `scheduled`, `site_visit`, `report_drafting`, `engineer_review`, `approved`, `closed`
`ticket.priority`	`low`, `medium`, `high`, `critical`
`ticket.closure.type`	`resolved`, `early`, or null
`account.role`	`admin`, `engineer`, `cs`, `technician`
`account.team`	`Team 1`, `Team 2`
`visit_day.status`	`pending`, `submitted`, `unfinished`
`checklist_item.status`	`pass`, `fail`, `na`, null
`checklist.template`	`residential`, `ci`, `ntp`
`report.status`	`draft`, `submitted`, `approved`
`report.format`	`residential`, `troubleshooting`
`maintenance_checklist[].result`, `report_notes.performance_report_result`	`accept`, `reject`, null
`summary.overall_status`	`good`, `fair`, `poor`
`site_improvement_performed`	`yes`, `no`, null
reading `remark`	`ok`, `not_ok`, null
`calendar_event.all_day`	true, false

12. Notes for the database design

- Replace names with ids for clients and accounts. Keep the display name as a separate column.
- Make `visit_days`, `messages`, `event_log` and the report parts child tables.
- Keep `event_log` append-only.
- Store the link between a ticket and an event once, not on both sides.
- Store timestamps as real date-time values in UTC, and show them in Malaysia time.
- Photos and files need real storage. Today only a label is saved.
- The "unfinished" day check should become a scheduled job.